Q3. File and directory management

Original question: Demonstrate basic file and directory management in Kali Linux. Create a directory, create a file inside it, navigate between directories, remove a file, and verify the results. Also display the current user and system hostname.

Aim

Create, navigate and remove a practice file, verify the outcome, and display the current user and hostname.

Requirements

Kali/Linux terminal, Bash and permission to write under your home directory.

Procedure

Step 1 - Identify the session and create a directory

```
whoami
hostname
mkdir -p "$HOME/EHF_Exam/Q3"
cd "$HOME/EHF_Exam/Q3"
mkdir Practice
ls -ld Practice
```

`whoami` displays the effective username. `hostname` displays the system's configured host name. `mkdir` creates Practice; `ls -ld` lists the directory itself rather than its contents. If Practice already exists, inspect it before reusing it.

Step 2 - Create a file and navigate

```
cd Practice
touch sample.txt
pwd
ls -l sample.txt
cd ..
pwd
ls -l Practice
```

`touch` creates an empty file when absent; on an existing file it updates timestamps. `cd ..` moves up one directory, so the second pwd should show Q3. The last command verifies the child directory still contains sample.txt.

Step 3 - Remove only the practice file

```
cd Practice
rm -i sample.txt
ls -la
cd ..
ls -ld Practice
```

Answer `y` only after verifying the named file. `rm -i` asks before unlinking it. `ls -la` includes hidden entries; `.` and `..` are normal. Practice should remain present and empty. Optional cleanup: `rmdir Practice` removes it only if empty, after you have recorded your result.

Expected output - illustrative

```
student
kali-lab
/home/student/EHF_Exam/Q3/Practice
/home/student/EHF_Exam/Q3
rm: remove regular empty file 'sample.txt'? y
# After deletion: sample.txt is absent; Practice remains.
```

User and host values vary. If you answer `n`, the file remains and the deletion task is not complete.

Observation and result

Record the actual user, hostname, before/after paths and final listing. The same file was created, verified and deleted while its parent directory remained available. This demonstrates relative navigation and targeted deletion.

Common errors

- Directory not empty: rmdir is deliberately limited; inspect contents instead of using recursive deletion.
- File still listed: check whether you declined rm's prompt or deleted a different path.
- Wrong directory: use pwd before any deletion. Linux filenames are case-sensitive.

Viva questions

- What does touch do to an existing file? It normally updates its access and modification times.
- What is an absolute path? A path starting at /, independent of the current directory.
- What is the difference between rm and rmdir? rm removes files; rmdir removes empty directories.
- Does hostname display the username? No; whoami displays the effective username.